HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 03 Name: _____ Marks: / 44

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The solution of $x - \frac{x-1}{2} > 5$ or $1 < x - 11$ is

   A. $x > 9$

   B. $x > 10$

   C. $x > 11$

   D. $x > 12$

(DSE-2013 PII Q05)

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2. Let $k$ be a constant. Solve $(x-k)^2 = 4k^2$.

A. $x = 3k$

B. $x = 5k$

C. $x = -k$ or $x = 3k$

D. $x = -3k$ or $x = 5k$

(DSE-2013 PII Q06)

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3. Let $f(x) = x^{13} - 2x + k$, where $k$ is a constant. If $f(x)$ is divisible by $x + 1$, find the remainder when it is divided by $x - 1$.

A. 0

B. $-1$

C. 2

D. $-2$

(DSE-2013 PII Q09)

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4. The actual area of a playground is $900\,m^2$. Its area on a map is $36\,cm^2$. Then the scale of the map is

A. 1 : 25

B. 1 : 50

C. 1 : 500

D. 1 : 250 000

(DSE-2013 PII Q12)

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5. It is given that $z$ varies directly as $x$ and inversely as $\sqrt{y}$. If $y$ is decreased by 64% and $z$ is increased by 25%, then $x$

A. increases by 20%.

B. increases by 80%.

C. decreases by 25%.

D. decreases by 75%.

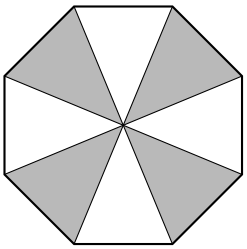
(DSE-2013 PII Q13)

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6. In the figure, the regular octagon is divided into eight identical isosceles triangles and four of them are shaded. The number of axes of reflectional symmetry of the octagon is



A. 2

B. 4

C. 8

D. 16

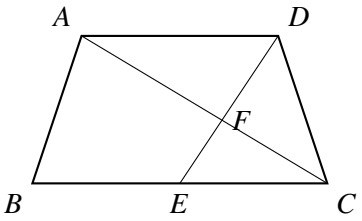
(DSE-2013 PII Q15)

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7. In the figure, $ABCD$ is a trapezium with $AD//BC$, and $AD : BC = 2 : 3$. Let $E$ be the midpoint of $BC$. $AC$ and $DE$ intersect at $F$. If the area of $\Delta CEF$ is $36\,\text{cm}^2$, then the area of the trapezium $ABCD$ is.




A. $216\,\text{cm}^2$

B. $264\,\text{cm}^2$

C. $280\,\text{cm}^2$

D. $320\,\text{cm}^2$

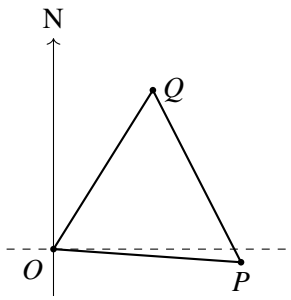
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8. In the figure, the bearing of $P$ from $O$ is $S86°E$ and the bearing of from $O$ is $N32°E$. If $P$ and $Q$ are equidistant from $O$, then the bearing of $P$ from $Q$ is




A. N24°W

B. N27°W

C. S24°E

D. S27°E

(DSE-2013 PII Q20)

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9. In $\triangle ABC$, $AB : BC : AC = 8 : 15 : 17$. Find $\cos A : \cos C$.

A. 8 : 15

B. 8 : 17

C. 15 : 8

D. 15 : 17

(DSE-2013 PII Q22)

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10. The coordinates of the points $A$ and $B$ are $(2,5)$ and $(4,-1)$ respectively. Let $P$ be a moving point in the rectangular coordinate plane such that $AP = BP$. Find the equation of the locus of $P$.

A. $x-3y+3=0$

B. $x-3y-7=0$

C. $x-3y+13=0$

D. $3x+y-11=0$

(DSE-2013 PII Q24)

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11. The equation of the circle $C$ is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points $P$ and $Q$ are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of $C$ is 5.

II. The midpoint of $PQ$ lies outside $C$.

III. If $G$ is the centre of $C$, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

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12. Two numbers are randomly drawn at the same time from seven cards numbered 1, 2, 3, 4, 5, 6 and 7 respectively. Find the probability that the product of the numbers drawn is an odd number.

A. $\frac{2}{7}$

B. $\frac{4}{7}$

C. $\frac{12}{49}$

D. $\frac{16}{49}$

(DSE-2013 PII Q26)

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13. $A00000E00001_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

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14. If $x - \log y = x^2 - \log y^2 - 10 = 2$, then $y =$

A. 100

B. 2 or $-4$

C. $\frac{1}{100}$ or 10000

D. $\frac{1}{10000}$ or 100

(DSE-2013 PII Q34)

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15. If $\alpha \neq \beta$ and $\begin{cases} 3\alpha = \alpha^2 - 5 \\ 3\beta = \beta^2 - 5 \end{cases}$, then $\alpha\beta =$

A. 3

B. $-3$

C. 5

D. $-5$

(DSE-2013 PII Q35)

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16. The real part of $i+2i^2+3i^3+4i^4$ is

A. 2

B. $-2$

C. 6

D. $-6$

(DSE-2013 PII Q36)

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17. Consider the following system of inequalities:

$$\begin{cases} x \geq 2 \\ y \geq 0 \\ x + 4y \leq 22 \\ 4x - y \leq 20 \end{cases}$$

Let $D$ be the region which represents the solution of the above system of inequalities. If $(x, y)$ is a point lying in $D$, then the greatest value of $3y - 4x + 15$ is

A. 3.

B. 17.

C. 22.

D. 30.

(DSE-2013 PII Q37)

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18. The $n$th term of a sequence is $2n - 19$. Which are true?

I. 25 is a term of the sequence..

II. The sequence has 10 negative terms.

III. The sum of the first n terms of the sequence is $n^2 - 18n$.

A. I only

B. II only

C. I and III only

D. II and III only

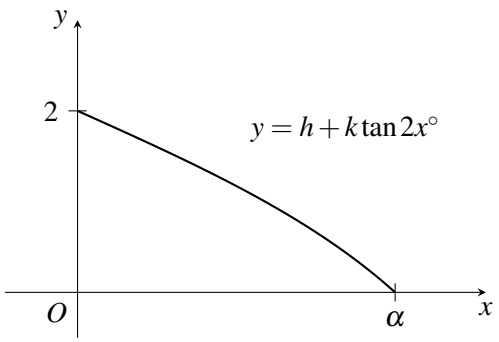
(DSE-2013 PII Q38)

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19. Let $h$ and $k$ be constants. The figure shows the graph of $y = h + k\tan 2x^\circ$, where $0 \le x \le \alpha$. Which of the following are true?




I. $h > 0$

II. $k < 0$

III. $\tan\alpha = \frac{1}{k}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

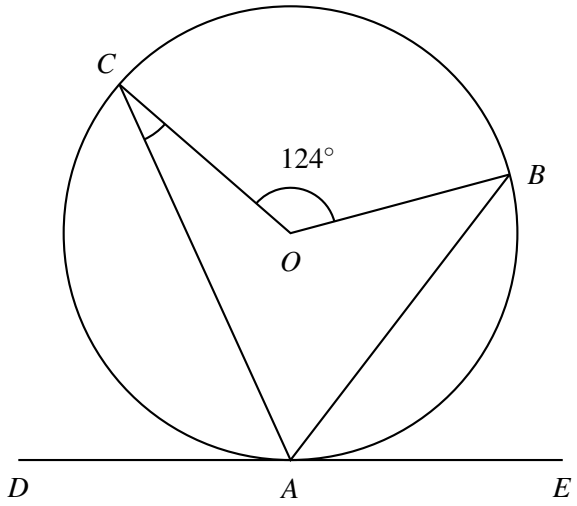
(DSE-2013 PII Q39)

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20. In the figure, $O$ is the centre of the circle $ABC$, and $DE$ is tangent to the circle at $A$. If $AB$ is the angle bisector of $\angle CAE$, then $\angle ACO =$




A. 26°

B. 28°

C. 31°

D. 34°

(DSE-2013 PII Q41)

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21. Let $O$ be the origin. If the coordinates of the points $A$ and $B$ are $(0, 12)$ and $(30, 12)$ respectively, then the $y-$coordinate of the circumcentre of $\Delta OAB$ is

A. 6.

B. 8.

C. 12.

D. 15.

(DSE-2013 PII Q43)

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22. If the variance of the five numbers $x_1, x_2, x_3, x_4$ and $x_5$ is 13, then the variance of the five numbers$3x_1 + 4, 3x_2 + 4, 3x_3 + 4, 3x_4 + 4$ and $3x_5 + 4$ is

A. 39

B. 43

C. 117

D. 121

(DSE-2013 PII Q45)

I can finish _____ questions

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